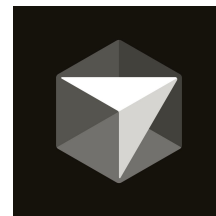
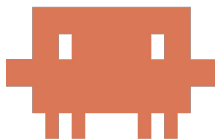
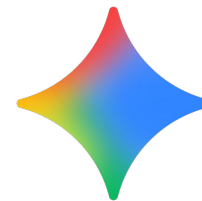


**Please
find and
sit next to
a partner!**



Software Engineering III

AI in Software Engineering



CS61B, Spring 2026

Guest Lecturer: Stella Kaval

Raise your hand:

- Used AI to explain code?
- Used AI to write tests?
- Used AI to debug?
- Used AI for a full feature or project?

If yes: was it mostly 👍 or 👎?



Introduction

CS61B, Spring 2026

Introduction

What does AI in software development look like?

What are the most important skills today?

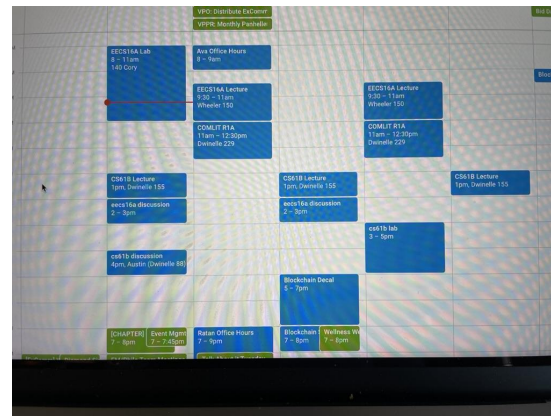
Q&A

Stella Kaval (she/her)
B.A. Computer Science 2025



Spring 2022, CS61B

- Office hours
- Favorite class at Berkeley!
- Became a TA, 3 years on course staff
- Guest lectures



```
1 setup2.inc
2 test01-init.in
3 test02-basic-checkout.in
4 test03-basic-log.in
5 test04-pre-checkout.in
6 test14-add-remove-status.in
7 test15-remove-add-status.in
8 test25-successful-find.in
9 test29-bad-checkouts-err.in
10 test32-file-overwrite-err.in
11
12 src
13   notwug.txt
14   wug.txt
15   Makefile
16   runner.py
17   tester.py
18   README.md
19
20 terminal: Local v. +
21 hasselements@116-25-167-299 proj3 A java gittit/Main init
22 hasselements@116-25-167-299 proj3 A
```

Hi Stella,

Here's your assigned seat for Midterm 2

Room: 10 Evans

Seat: D14

You can view this seat's position on the map at <https://seating.cs61b.not.cs61a.org>

CS 61B's Midterm 2 will take place

Summer 2023, Oracle

6 weeks to onboard

- Knowledge scattered
- Codebase huge
- Blockers

6 weeks coding (feedback tool)

- No AI tools
- New language (React)
- All code written line by line
- ~1000 lines of code



Summer 2024, Microsoft

2 weeks to onboard

- Internal Copilot tool answers questions immediately

10 weeks coding (datacenter buildout tools)

- GitHub Copilot
 - Generated 20-50 line chunks
 - Mainly frontend
 - Error prone
- ~2000 lines of code
 - Finished project early
 - Led an additional accessibility project



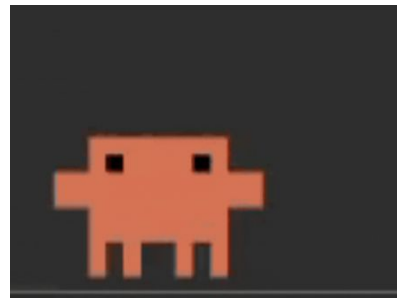
2025, Meta

- Similar speedy onboarding
- LLMs can code confidently (GPT)
 - 100+ line chunks
 - Across multiple files, functions, classes
 - Never wrote a single test!
 - Still struggling with C++, longer tasks, debugging, hallucinations
- By 12 weeks (span of internship)
 - ~20,000 lines of code
 - 80% AI generated code



2026, Meta

- "Vibe coding" (**What is vibe coding?**)
 - Claude Code (Opus 4.6)
 - AI can handle full features (1000+ lines)
 - Runs terminals and commands
 - Sends messages, write docs, comments
 - Deep data analysis and debugging
- Today
 - Parallelize many projects/tasks/investigations
 - Automated repeated tasks
 - Overnight agentic flows
 - Role changed (product manager, engineer, data analyst, customer support)
 - Taking up projects we never thought possible



What changed and what stayed the same?

- What changed?
 - How fast we write code
 - Tools used
- Stayed the same
 - Understanding what you're building, why you're building it, and who it's for
 - Trade offs and constraints (runtime, data structures, architecture)
 - Code and design review
 - Collaboration



Business Insider

<https://www.businessinsider.com> › Tech › AI ⋮

Meta's AI Push Ties Employee Goals to AI Tool Adoption

Mar 26, 2026 — One target is for 80% of mid to senior-level **engineers** to adopt **AI** tools such as DevMate, **Metamate**, and Google's Gemini, **with** a note that the ... [Read more](#)



The New York Times

<https://www.nytimes.com> › Podcasts › The Daily ⋮

The Workers Letting A.I. Do Their Jobs

Apr 14, 2026 — In an era of agents powered by **artificial intelligence**, many **programmers** are barely programming.



Business Insider

<https://www.businessinsider.com> › Tech › AI ⋮

Google Says 75% of the Company's New Code Is AI- ...

5 days ago — **Three-quarters of new code created inside Google is now generated by AI** and reviewed by human engineers, the company said Wednesday. [Read more](#)



TechCrunch

<https://techcrunch.com> › 2026/04/17 › tokenmaxxing-i... ⋮

'Tokenmaxxing' is making developers less productive than ...

Apr 17, 2026 — Now, the company is releasing new tools that track the metadata generated by **AI** agents, offering analytics on the quality and cost of their code ... [Read more](#)

What does AI in software development look like?

CS61B, Spring 2026

Introduction

What does AI in software development look like?

What are the most important skills today?

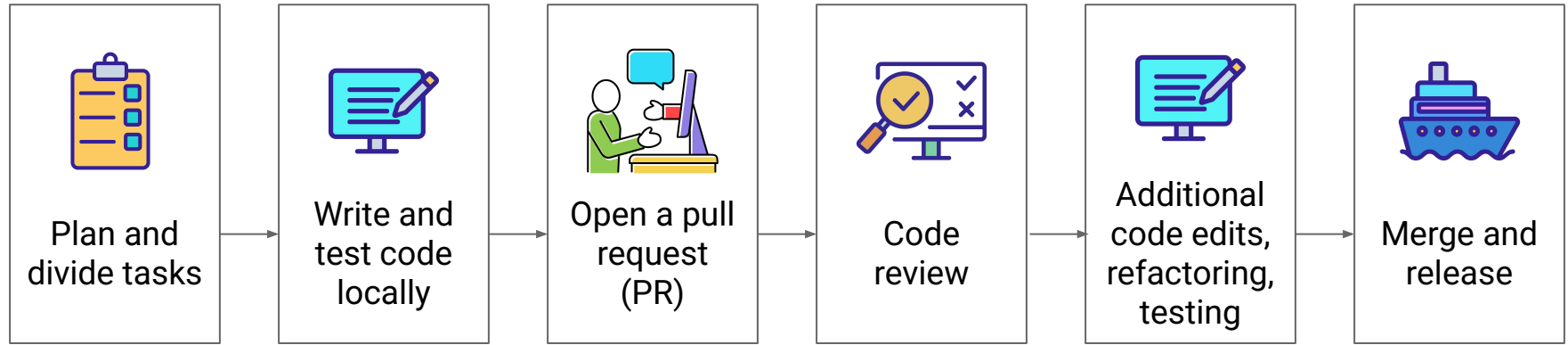
Q&A

What is software engineering?

What is software engineering?

- ChatGPT
 - Software engineering is the process of designing, building, testing, and maintaining software in a reliable and organized way.
- Claude
 - Software engineering is the disciplined application of engineering principles to design, build, test, and maintain software systems.
- Gemini
 - Software engineering is the systematic, disciplined, and quantifiable application of engineering principles to the design, development, maintenance, and retirement of software to ensure it is reliable, efficient, and scalable.

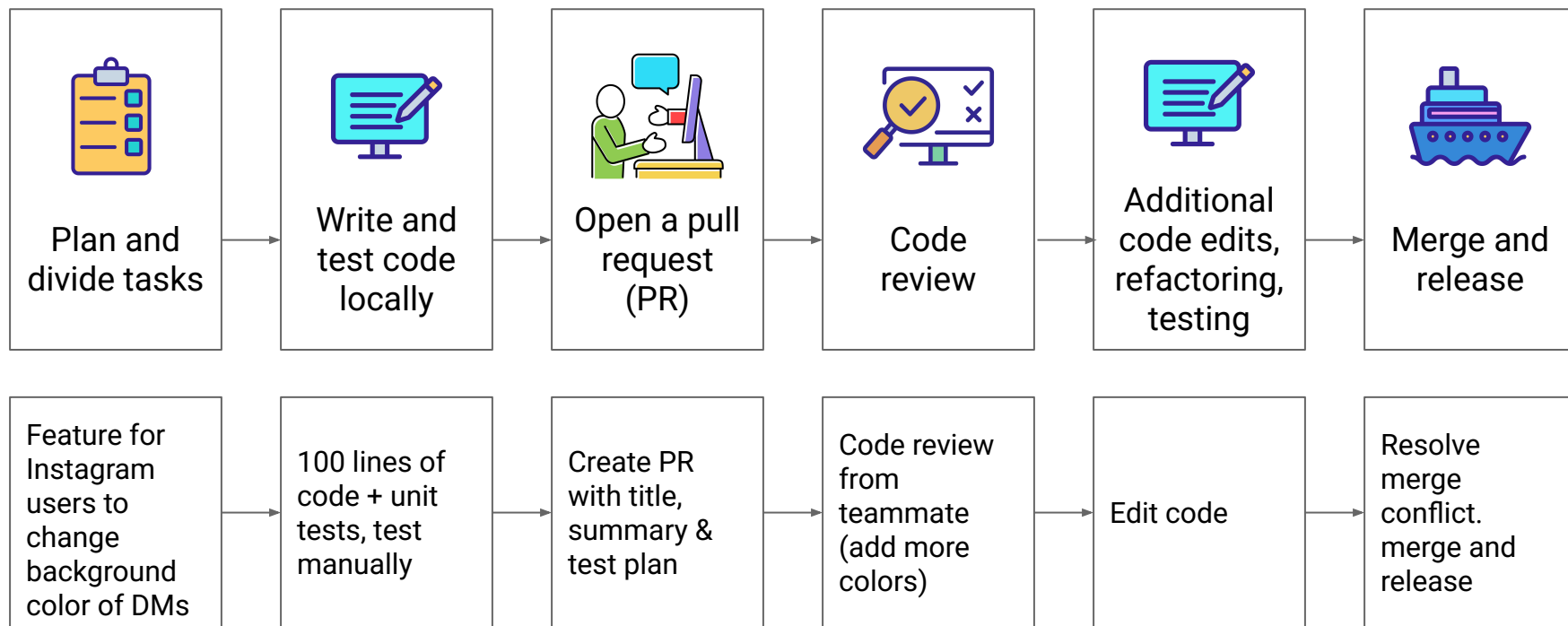
Software Development Cycle



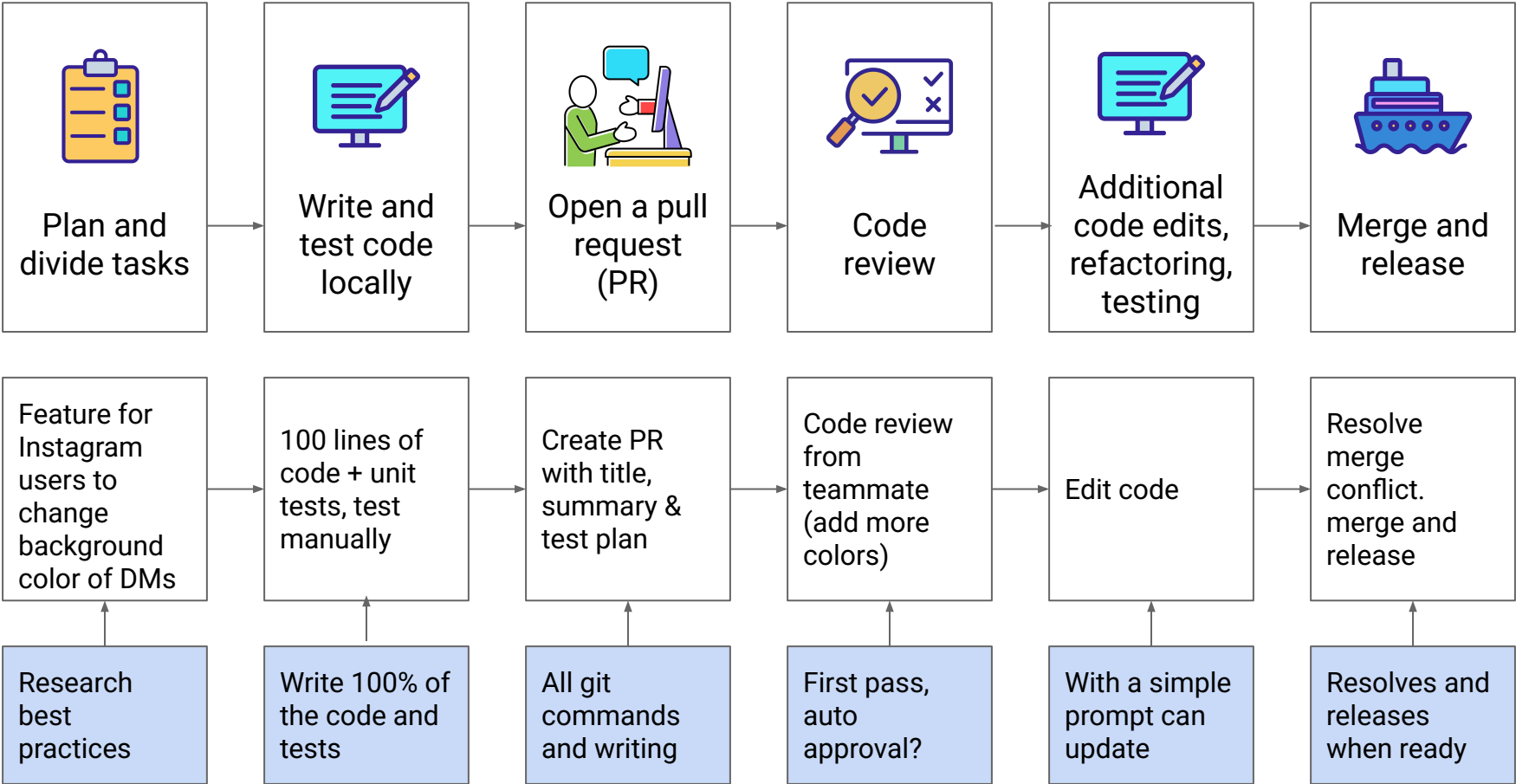
Today's example

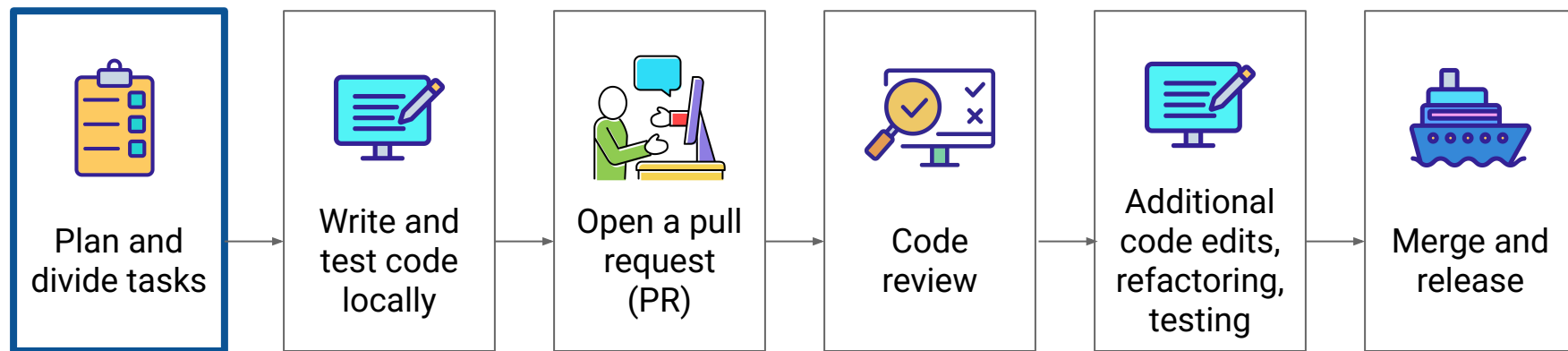


Software Development Cycle Without AI



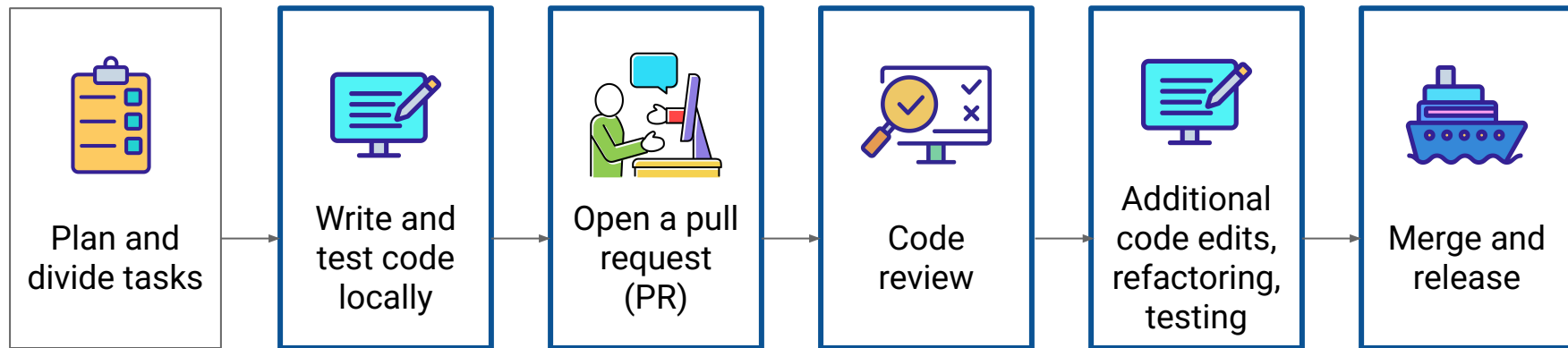
Software Development Cycle With AI





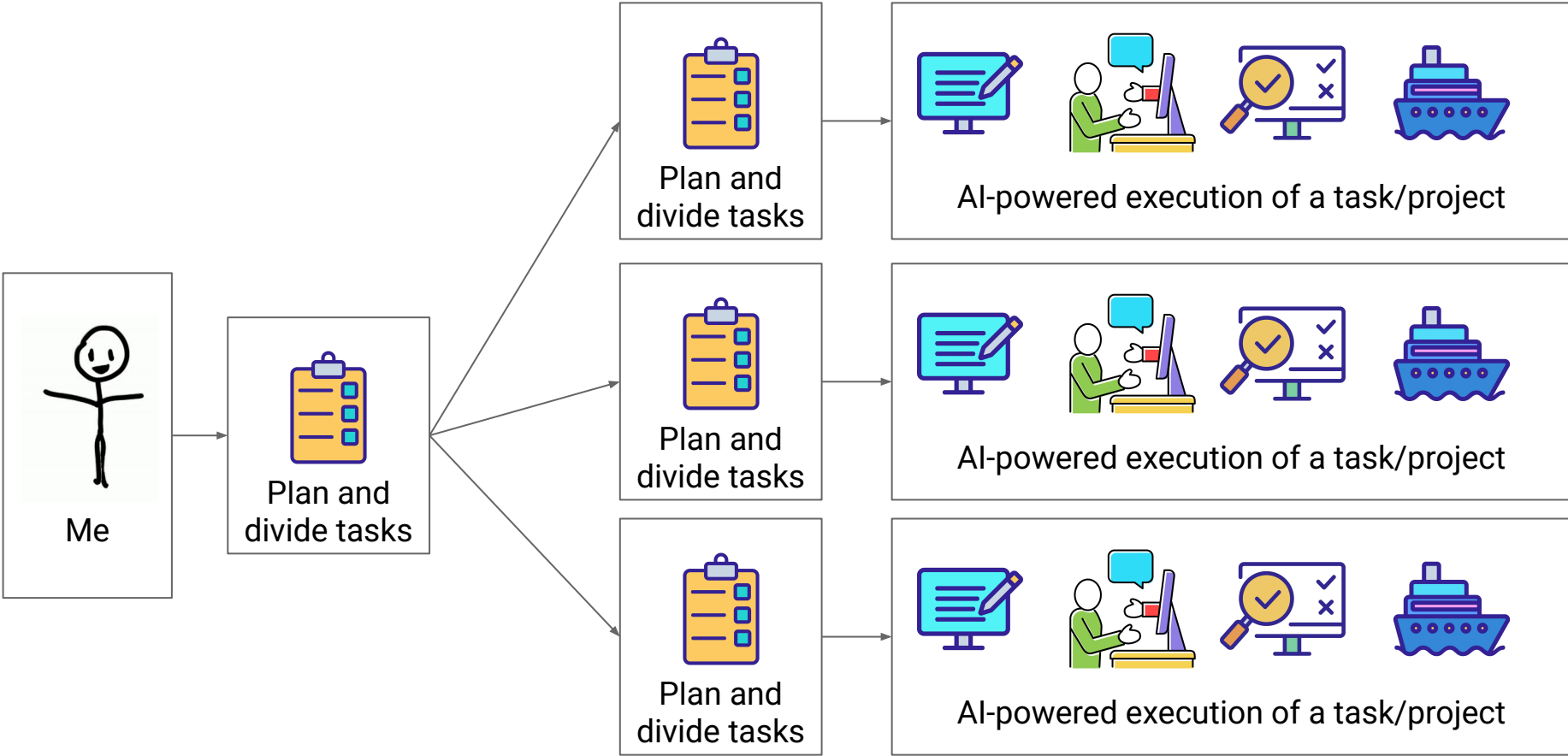
- Key shift: writing code is becoming cheaper, so planning matters more than ever
- If you give AI a vague or bad spec, it can generate a lot of bad code
- A good spec makes AI more useful because it tells the model what matters, what does not matter, and how the work will be judged
- The engineer's value moves upstream into defining the right problem and downstream into verifying the result

Software Development Cycle

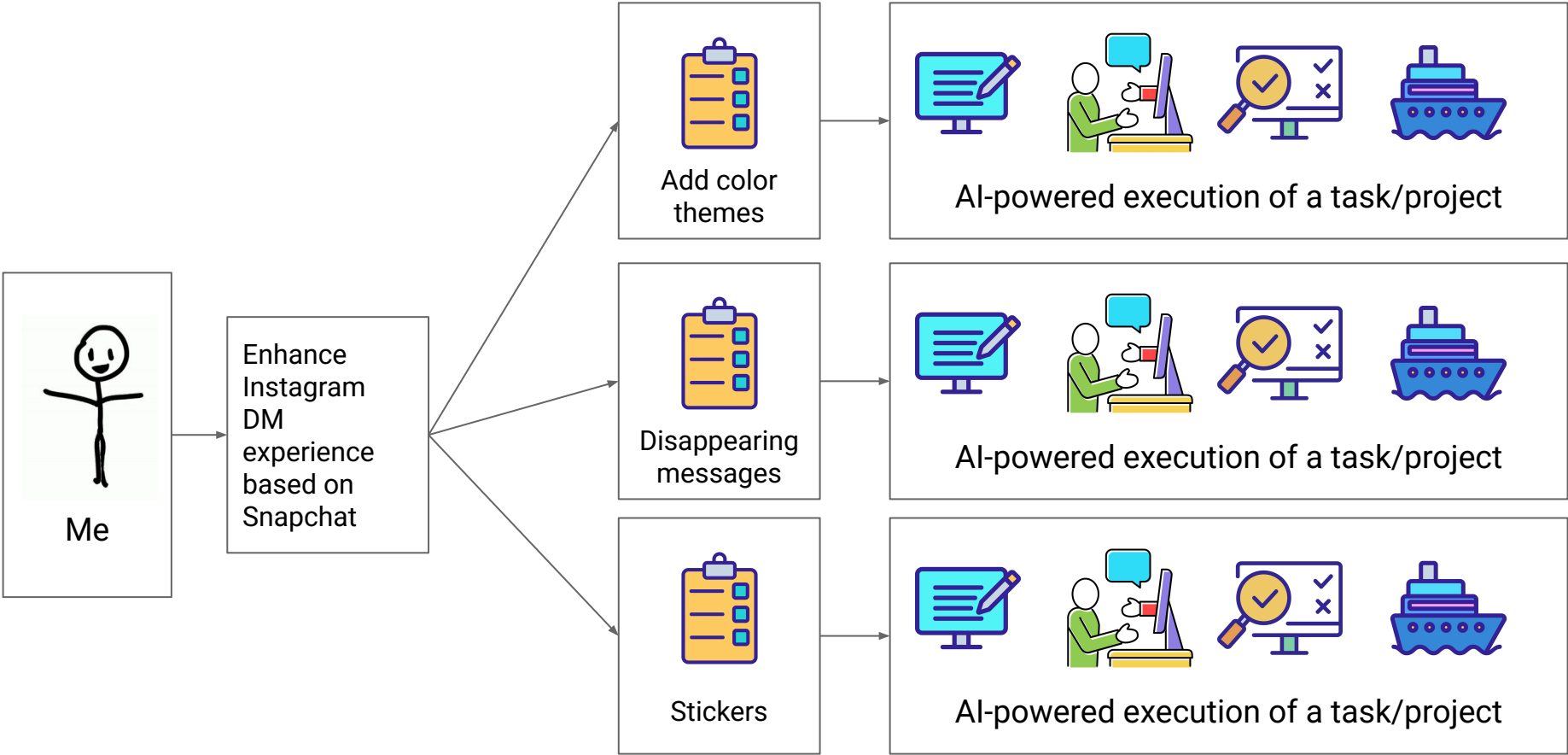


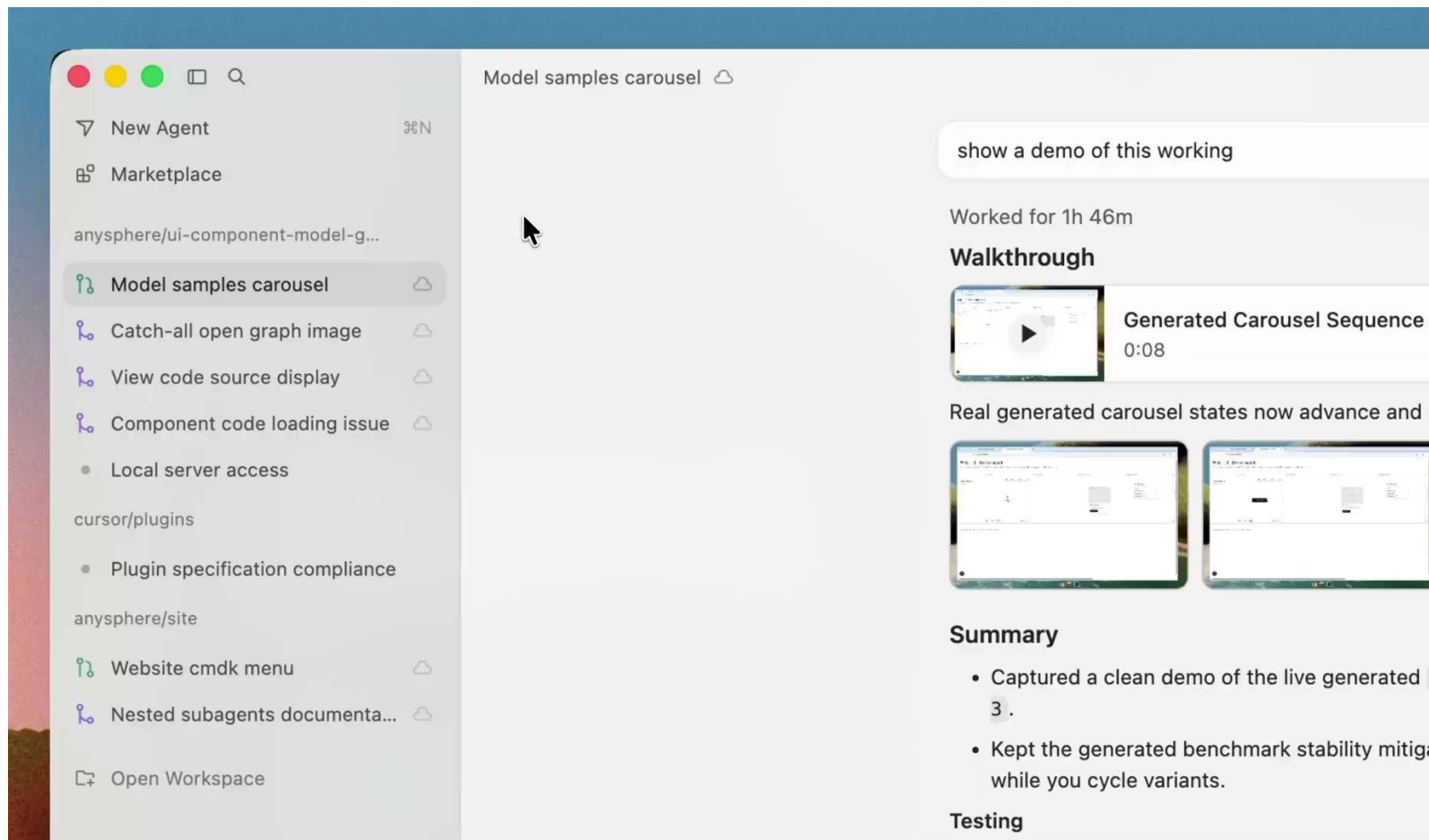
- AI can accelerate the actual building phase, and the tools are going to keep improving
- How do you use AI so it produces something small, reviewable, and aligned with your goal? (more soon)

My AI Software Development Cycle



My AI Software Development Cycle





The screenshot displays the 'My AI Software Development Cycle' interface. On the left is a sidebar with a search bar and a list of project links. The main content area on the right features a 'Model samples carousel' section. This section includes a text input 'show a demo of this working', a status 'Worked for 1h 46m', a 'Walkthrough' heading, a video player for 'Generated Carousel Sequence' (0:08), and a description 'Real generated carousel states now advance and'. Below this is a 'Summary' section with two bullet points and a 'Testing' section.

Sidebar Links:

- New Agent
- Marketplace
- anysphere/ui-component-model-g...
- Model samples carousel**
- Catch-all open graph image
- View code source display
- Component code loading issue
- Local server access
- cursor/plugins
- Plugin specification compliance
- anysphere/site
- Website cmdk menu
- Nested subagents documenta...
- Open Workspace

Main Content Area:

Model samples carousel

show a demo of this working

Worked for 1h 46m

Walkthrough

Generated Carousel Sequence
0:08

Real generated carousel states now advance and

Summary

- Captured a clean demo of the live generated 3.
- Kept the generated benchmark stability mitigated while you cycle variants.

Testing

What are the most important skills today?

CS61B, Spring 2026

Introduction

What does AI in software development look like?

What are the most important skills today?

Q&A



Design



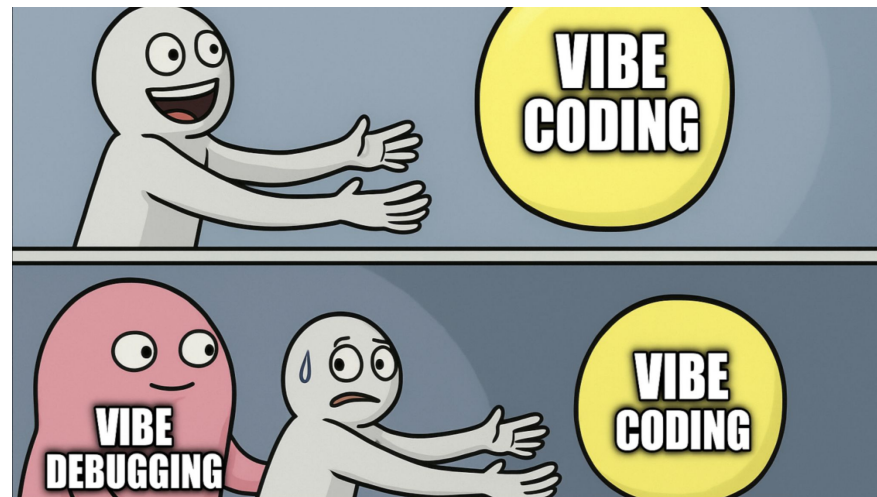
Coding with AI

If AI could generate a login page for a website in 10 seconds, why do we still need you to plan it?

- AI can generate code quickly
- Humans decide the system, constraints, and tradeoffs
- The hard part is not “can AI build it?”
- The hard part is “what should it build, and is it correct?”

Why planning matters

- Vague prompt → lots of wrong code
- Clear design → smaller, reviewable changes
- A design doc becomes context for both humans and AI
- Reduces wasteful AI slop you would probably end up discarding



What goes in a design doc?

- TLDR, stakeholders, and impact
- Goals and non-goals
- Alternatives considered
- Architecture and data model
- Test plan: unit, integration, end-to-end
- Implementation plan
- Rollout and rollback plan



How to Think Like a Software Architect in the AI Era Example

TLDR, stakeholders, and impact

- Let Instagram users choose approved DM themes, measuring # times changed to increase how long users stay on app

Goals and non-goals

- Support accessible preset themes per DM thread; do not allow custom uploads or color pickers

Alternatives considered

- Considered global themes and custom colors, but preset per-thread themes are safer and simpler for version 1

Architecture and data model

- Store a validated theme_id on each DM thread and map it to approved theme assets on the client
- Coordinated with client assets team and they approved the extra metadata

Test plan: unit, integration, end-to-end

- Test valid themes, saving/loading across devices, and manually verify the full DM experience.

Implementation plan

- Add theme metadata, backend save/load support, theme picker UI, and updated DM rendering.

Rollout and rollback plan

- Launch to a small user group first and fall back to the default theme if issues appear.

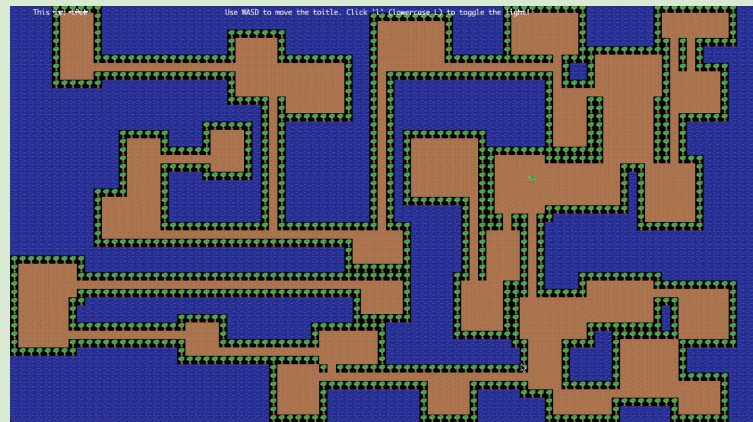
Exercise: find a partner!

Create a design document for a new feature in your CS61B BYOW game

- 4 minutes
- < 1 page, bullet point format
- Use any AIs tool you want

<https://tinyurl.com/designdoc61b>

<https://tinyurl.com/designdoc61bsubmit>





Design



Coding with AI

Context: the model's working memory in the chat

Context window: the maximum amount of text the model can use at once

Tokens: small chunks of text the model reads and writes

Skills: reusable task-specific instructions, resources, or workflows

Agents: AI systems that can plan, use tools, observe results, and continue

Visual Idea for Context

- AI is like a student walking into office hours
- The backpack = what it has in context
- Put in the spec, relevant files, examples, and constraints
- Leave out unrelated stuff
- A huge backpack with random stuff can still make the student's understanding worse



	✓ Good context	✗ Bad context
 What it is	✓ Clear, relevant info. <i>Ex: "Build a settings page. Here are the requirements..."</i>	✗ Missing, unclear, or overwhelming. <i>Ex: "Here's the entire project."</i>
 Purpose	✓ Provides clarity and direction. <i>Ex: "Help implement pagination for the user list."</i>	✗ Leads to confusion and low-quality results. <i>Ex: "Just fix it."</i>
 Examples	✓ Includes key context. <i>Ex: "Design doc, files, example style, what not to change."</i>	✗ Missing or unhelpful context. <i>Ex: "Old attempts." "Vague 'fix this' prompts."</i>
 Impact	✓ Saves time and gets better results. <i>Ex: "Clear context = faster, better results."</i>	✗ Wastes time and causes frustration. <i>Ex: "More confusion = more delays."</i>
 Key takeaway	✓ Give just enough context: clear, specific, actionable.	✗ Avoid too much, too little, or unhelpful context.

- **Describe bugs precisely:** what is wrong, where it happens, and what the expected behavior should be.
- **Use existing code as an example;** in industry, you often want AI to follow team helpers and patterns instead of inventing new ones.
- **Coordinate edits across files** by naming where to edit, what to import, and what should not change.
- **Use a living Markdown plan** so AI can track progress, decisions, and remaining tasks.

Let's work on revising some prompts!

Bug scenario: Player loads a saved world → screen is blank

Weak prompt: *"My world doesn't render after loading, fix it"*

Bug scenario: Player collects coins but score never updates or is shown anywhere

Weak prompt: *"Fix the coin feature"*

Attendance!



- **Even if you have a great prompt, AI can lead you astray**
 - Context window gets messy
 - It hallucinates APIs or commands
 - It repeats the same bug
 - It changes approach randomly
 - It crashes or loses track
- **How to recover:**
 - Stop and start over
 - Shrink the task
 - Be the AI's TA
 - Ask yourself: do we need to give AI some tool that they don't have yet?



AI IDE after I send "Still broken" for the 15th time



If AI writes all your code and AI writes all your tests, should you send it straight to production?

What matters:

- End-to-end behavior (does it actually work for users?)
- Human explanation of why the change is needed
- Multiple reviews (another AI or teammate)
- Correctness, security, performance, readability, test coverage

AI generates code. Your job is to judge and defend it.

Actionable Takeaways

1. [Github Copilot](#) (free with .edu email)
 - Integrated to VSCode you are already familiar with
2. [Cursor](#) (free)
 - New AI-first IDE
 - Can run any model on it
3. [Claude Code](#) (free tier available)
 - What I use at Meta

Q&A

CS61B, Spring 2026

Introduction

What does AI in software development look like?

What are the most important skills today?

Q&A

Practice using AI tools as often as you can, but use them thoughtfully

stellakaval@berkeley.edu

